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Assessment of charging technologies, infrastructure and charging station recommendation schemes of electric vehicles: A review

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ABSTRACT

This article comprehensively reviews recent Electric Vehicle (EV) charging infrastructure, technology, and issues related to charging station identification. A literature study on the above key areas shows an increasing consideration of EVs over the past decade. Further, the recommendations for user-friendly selection of the Charging Station (CS) are discussed. The transport sector is one of the leading contributors to global warming. It has a high dependence on fossil fuels such as petrol and diesel, which are expected to be scarce shortly. Therefore, EVs have attracted considerable attention as they can minimize fuel consumption and emissions. For the last few years, studies related to Plug-in-Hybrid Vehicles (PHEVs)/EVs have drawn the researcher's attention. Moreover, several technical articles associated with charging technologies with available infrastructures are discussed, and we highlighted their pros and cons, particularly in charging infrastructure, and the charging power and power transfer between EV and grid are broadly discussed. From a customer perspective, the selection of charging stations and associated problems are presented. Finally, the suggestions for the user-friendly selection of charging stations are discussed.

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1. Introduction

In recent years, many countries have implemented regulatory requirements to solve issues in the transport sector, such as escalating the need for non-renewable fuels and minimizing pollution in metropolitan areas [1,2]. Thus, great attention has been given to eco-friendly EVs due to concerns about the availability of fossil fuels, the environment, and emissions regulation [3–5]. Few countries have stipulated significant guidelines to ease EVs' expansion and adoption [6,7]. For example, several cities in North America are in the changeover period to rapidly replace conventional vehicles with PHEVs and then with EVs [8,9]. EVs have significant benefits in terms of cost and convenience [10]. Unlike internal combustion engine (ICE) vehicles, maintenance costs do not exist in an EV, including the cooling liquid, starter battery, lubricant, and exhaust system that carries out with time [11]. In general, EVs are classified as Battery-powered EVs (BEVs), Hybrid EVs

Abbreviations: BEV, Battery-powered electric vehicles; CS, Charging station; DCFC, Direct current fast charging; EPRI, Electric power research institute; EV, Electric vehicle; EVSE, Electric vehicle supply equipment; FCS, Fast charging station; FIFS, First in first serve; HEV, Hybrid electric vehicles; ICE, Internal combustion engine; IEC, International electro-technical commission; IoT, Internet of things; JCR, Journal citation reports; PHEV, Plug-in-hybrid vehicle; PL, Parking lot; RES, Renewable energy source; SAE, Society of automotive engineers; SOC, State of charge; ToU, Time-of-use; V2A, Vehicle to anything; V2G, Vehicle to grid; V2B, Vehicle to building.

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(HEVs), and plug-in HEVs (PHEVs). A BEV solely depends on a rechargeable battery for propulsion because it does not have an ICE [12]. The ICE powers the HEV, and regenerative braking charges the battery. The PHEV also operates similarly to PEV but can charge the battery from external charging points. The driving range of the BEV depends on the battery's capacity, whereas the PEV and PHEV driving range dependent on both ICE and the battery [13].

The BEV battery range can vary from a limited range (Nissan Leaf-24 kWh) to a long-range (Rimac C II-120 kWh) vehicle, and the HEV battery has a capacity of 0.9 kWh (Toyota Yaris Hybrid) to 1.78 kWh (Toyota Pirus I). The PHEVs battery capacity varies from 4.4 kWh (Toyota Prius III) to a maximum of 34 kWh (Polestar 1) [14]. The volume of EV charging, location, and time are random. Therefore, many EVs may radically affect the electric power network, owing to the elevated power utilization. A few of the most acute crises in distribution systems are associated with non-desired energy utilization peaks, overloading of transformers and augmented power loss, and others [1,12,15–17]. It also affects the stability of the grid [18,19].

Consequently, several studies have been conducted on EVs around the world. In the current research study, the researchers mainly aim to deliver an extensive overview of EV charging technologies, different optimization problems and computational methods to solve problems associated with charging/discharging EVs and the challenges involved with the charging infrastructure to adopt EVs in the market. Thus, a literature survey was conducted through the database of journal citation reports (JCR). The survey covers the duration from 2010 to 2020. Fig. 1 presents EV articles related to three major domains: charging technology, charging infrastructure, and charging station recommendations. It illustrates the rapidly growing research trend of EVs.

The rest of the paper is organized as follows: In Section 2, EV charging technologies are presented. The charging infrastructure is presented in Section 3, and the charge station recommendation schemes are described in Section 4. The conclusion is given in the last section.

2. EV charging technology

An overview of different power levels, costs, merits, and demerits of various charging technology is discussed. The charging time of an EV depends mainly on the charging level of the battery. Besides, it also depends on the ability of the EV battery to accept a high charging rate, the cable used for charging, and the Electric Vehicle Supply Equipment (EVSE). EVSE is the "point-of-fuelling" infrastructure used to deliver electrical energy to the charger, also called an electric re-charging unit [20]. It consists of various components such as charging cords, ports, connectors, and interfaces to charge the battery. Different charging modes/levels are classified by the international standard IEC 61,851 [21], the Electric Power Research Institute (EPRI), the Society of Automotive Engineers (SAE) [22], and the International Electro-Technical Commission (IEC) [23], namely alternating current AC - Level 1, Level 2, Level 3, and Direct Current DC level 4 [24].

The EPRI report disclosed that EV users are willing to charge their EVs after arriving home, particularly at night [25]. Thus, AC level 1 (slow) charging is preferred for EV charging at home [26] and is called overnight or residential. Further, the slow charging equipment is suitable for fixing typical wall outlets at home. In the level 1 charging mode, electric power is converted from AC to DC by the vehicle's onboard charger [27]. However, this power conversion may not be the same in every region/country because of various standards and frequency.

2.1. Level-1 charging

In North America, a standard 120 V/15 A single-phase outlet grounded, such as NEMA 5-15R, is used as a level-1 charging, and this connection operates with the standard SAE-J1772 connector [28]. The current handling capacity of this mode has two options, either 15 A (12 A usable) or 20 A (16 A functional), and it can draw from 1.4 kW to 1.9 kW of power. This mode will take an average of 8–12 h to reach 100 % battery State of Charge (SOC). The advantage of the level-1 charging mode is that no additional infrastructure is required for homes and office sites; hence it is economical. The total cost of a level-1 charging infrastructure has been reported as approximately 400–900 USD [29]. However, support for communication control is not available. So, it could not consider the grid load and load quality when connected to the grid. As a result, many level 1 charges could negatively impact the grid, such as grid power congestion.

2.2. Level-2 charging

Level 2 charging mode is the most suitable option for private charging facilities. For private installation, this mode assigns a single-phase 240 V AC with a current-handling capacity of 40 A and 80 A for a 400 V (three-phase) AC supply [30]. Because of standardized vehicle-to-charger connection and less charging time than mode-1, EV users are likely to prefer mode-2 charging technology. A residential infrastructure installation of mode-2 technology reportedly costs approximately 2,150 – 2,300 USD [31]. Furthermore, the quickest car in the world "Tesla Roadster" charging system, is reported to add a cost of 3,000 USD [32]. The mode-2 charging is advantageous, as the supplied equipment and the office/warehouse appliances can operate with the same circuit (i.e., electric ovens) [33]. Users can utilize a "temporary-use cable" provided by the supplier to charge EV batteries outside their homes. It can communicate with the car and can contribute toward grid stability. The cable can protect itself from the issues such as overcurrent and over-temperature [34]. This charging mode usually takes 4–8 h to charge the battery, which is better than mode-1 charging. Commercial and public charging stations cost more than 15,000 USD. Thus, the infrastructure installation cost is cheaper than the commercial charging infrastructure [35]. Though mode-2 charging has many advantages, it has certain drawbacks, such as the power consumption can surge up to 25 % [36].

2.3. Level-3 charging

A few years ago, SAE developed a fast-charging option to charge an EV battery capacity of 130 kW, using 480 V(AC) with a considerable quantity of high current [37], standard in commercial and industrial locations. Therefore, highway rest areas and city refueling points, such as malls, airports, parks, motels, petrol pumps, and other public places, are suitable for installing level 3 charging. It can enable the EV user to charge the EV while performing other tasks [38]. A Japanese protocol for "CHAdeMO" (move using charge) is gaining international recognition for quick charging via a special electrical connector. In the EU, Combined Charging System connectors are recommended to install in Fast Charging Station (FCS). As per the EU regulation, a type 2 (IEC 62196) DC connector is used in CCS.

As the level 3 chargers can support a high-power output, the chargers are much bulky and heavier. By using the level-3 charging option, EVs can be charged very quickly. But it has considerable drawbacks;

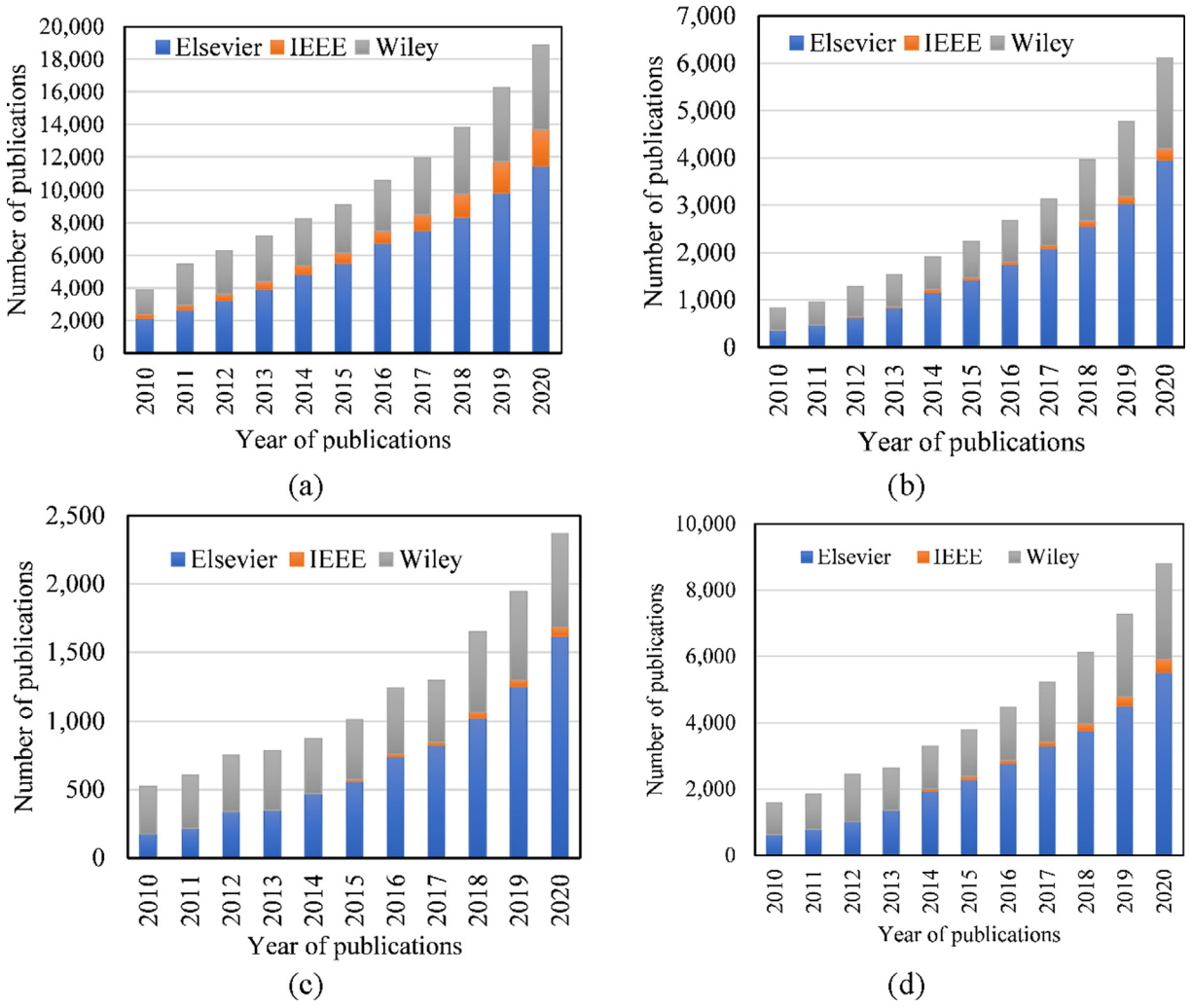


Fig. 1. Published research articles from 2010 to 2020 on (a) electric vehicles, (b) EV charging management, (c) charging technology, and (d) charging infrastructure.

1. The level-3 charging option has significantly less impact in the residential area.
2. Level 3 charger requires a dedicated cooling module for high-power electronics.
3. The cost of installation for a level 3 charging infrastructure is very high. It costs between 30,000 USD and 160,000 USD, depending on the quality.
4. The DC plugs and hardware standards are in the improvement stage [39].
5. The maintenance of the CS is another cost factor that cannot be neglected [40].

In the end, high-power charging will increase the power demand, and it has the potential for a sudden overload in local distribution equipment at peak hours [41,42].

2.4. Level-4 charging

The DC charging facility is mainly required for long-distance driving and heavy vehicles and is unsuitable for residential places [43]. For instance, a typical Electrify America highway charging station with two 350 kW and four 150 kW chargers will increase the power consumption by 1.2 MW, necessitating the purchase of a pricey distribution transformer that has been improved [44]. In order to support the uncontrolled charging behavior of the significant number of EVs anticipated to be in use by 2040 in Norway, it is predicted that $1.6 billion will be needed for grid strengthening [45]. The voltage level of a DC fast charging is up to 800 V. One of the main advantages of DC fast charging is that the charging time is less than the other charging levels [46]. However, the cost associated with this charging is high. 80,000 USD to 120,000 USD is required to install a Direct Current Fast Charging (DCFC), including investment, infrastructure, and O&M costs [47,48]. It requires a high-level protection system in case of a fault occurs. In addition, the DCFC station must be modified significantly for different EV cases. The DCFC is an emerging area, and many research works are being carried out. Table 1 provides the different charging stations for EVs followed by SAE standard J1772. Table 2 shows other EVs' battery capacity and their battery chemistry.

3. Charging infrastructure

The development of charging infrastructure has gained significant attention in developed and developing countries as inadequate charging infrastructure can affect EV promotion in the market. Hence, academicians and research & development (R&D) engineers are more willing to create a state-of-the-art charging infrastructure [52–55]. The charging infrastructure comprises power and a communication system [56].

Table 1
Charging stations schemes [SAE standard J1772 [43,49,50].

Power level	Voltage and current	Location of charger	Typical use	Max. power (kW)	Vehicle technology	Possible to control charging	Charging time (hrs)	Energy supply interface
1	120 V_{ac}-15 A, 120 V_{ac}-20 A	1ϕ– On-board	Home or office	1.4 1.9	PHEVs (5–15 kWh)EVs (16–50 kWh)	No	4–11 11–36	Convenience outlet
2	240 V_{ac}-40 A, 400 V_{ac}-80 A	1-phase/3-phase- On-board	Private	4 8 19.2	PHEVs (5–15 kWh)EVs (16–30 kWh)EVs (3 –50 kWh)	Yes	1–4 2–6 2–3	Dedicated EVSE
3	208 V_{ac}-80 A, 200A 600 Vac-80A, 200A	3-phase- -Off-board	Public outlets	38.4 130	EVs (20–50 kWh)	Yes	0.4–1 0.2–0.5	Dedicated EVSE
4	400–800 V DC	Off-board	Highway	50–500*	EVs (50 kWh and above)	Yes	0.1–0.2*	Dedicated EVSE

* Based on the manufacturer.

Table 2
Battery capacity and chemistry for EV charging from different manufacturers [51].

Serial	Model	Company	Vehicle type	Battery size (kWh)	Battery chemistry	Energy available (kWh)
1	Leaf	Nissan	EV	24	Li-ion (LMO)	19.2
2	i-MiEV	Mitsubishi	EV	16	Li-ion (LMO)	12.8
3	RAV4	Toyota	EV	42	NiMh	33.6
4	Cooper	BMW	EV	28	Li-ion	22.4
5	E6	BYD	EV	75	Li-ion	60
6	Stella	Sabaru	EV	9.2	Li-ion	7.36
7	Roadster	Tesla	EV	53	Li-ion	42.4
8	Chevrolet Spark	General Motors	EV	21.3	LFP	17.04
9	500e	Fiat	EV	24	(LMO-NMC)	19.2
10	Chevrolet Volt	General Motors	PHEV	17.1	(LMO spinel)	13.7
11	Prius	Toyota	PHEV	5.2	(NCA)	4.1
12	karma	Frisker	PHEV	22	Li-ion	17.6

3.1. Charging power

Depending on the charging power, the charging system is classified into residential charging, public charging, and charging at the workplace, warehouse, private, or remote charging system [57]. Residential charging is a widespread option among EV owners residing in detached houses, rowhouses and apartment buildings with the access to park and charge EVs on their charging facility. A residential charging system is a simple charging system that delivers electricity to an EV at low voltage. This system does not transmit any of the charging information to an external party; hence, it is neither controlled nor monitored [29]. Though the residential charging system's charging capacity is low, much research has been done on the home charging system to benefit the market participants such as users, aggregators, utilities, and the like [58,59,68–77,60,78–85,61–67].

In a workplace EV charging system, the flexible operation is carried out by regulating EV charging as the charging power level increases [78]. A large number of EVs parked in the Office Parking Lot (PL) could reduce the stress on the grid through the vehicle to grid mode, and it can reduce the office's dependence on the energy from the grid through the vehicle to building [78,84–88].

Public charging is mostly consisting of fast and level 2 chargers. Many customers have a concern about the charging power and waiting time. Thus, a public charging station is more suitable for potential customers. However, the location and placement of public CS are to be carefully selected as the improper placement of FCS could negatively impact the grid and profit of the CS operators [89–93]. Some of the research works presented the potential of renewable energy in EV charging. In general, by integrating the Renewable Energy Sources (RESs) with the PL/CS, several benefits are achieved, such as 1) Minimizing the energy consumption of PL/CS from the main grid by participating in a demand-side management program. 2) Minimizing the grid peak load during peak hours 3) Minimizing the additional cost investment for power generation 4) Maximizing the profit of the PL/CS through Vehicle to Grid (V2G) and Vehicle to Building (V2B) mode. 5) Reducing the yearly net greenhouse gas emission and operational cost [24,39,44,47,49,55,62,63,65–68].

3.2. EV and grid power transfer

Proper management of charging power would benefit all the participants involved in EV charging. It will maintain the grid stability and ensure the power balance between the demand and supply. As discussed in the previous section, EV charging could occur from different locations, and proper scheduling is desirable to control the charging power. Consequently, the inputs from EV, CS, grid, and an aggregator are needed for optimization to meet the objectives such as minimizing cost, charging time, and waiting time. Hence, it is necessary to understand the relationship between EVs and the grid. A generalized EV charging system is presented in Fig. 2. The power flow between EV and grid can happen in two modes: unidirectional and bidirectional. Accordingly, the charging management is classified into coordinated and uncoordinated EV charging.

3.2.1. Uncoordinated EV charging

In uncoordinated EV charging, all individual EV owners have a charging schedule, and the need for an agent or operator is neglected. Hence, the users can reduce their EV charging cost by charging their EV whenever the electricity price is low. Most unco-

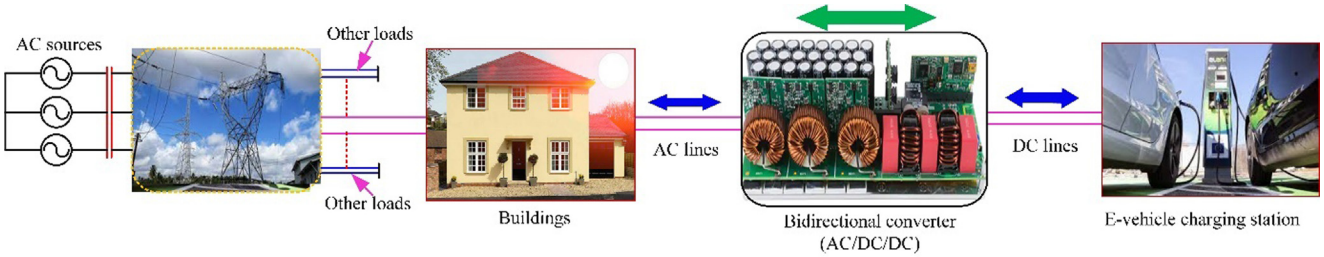


Fig. 2. Generalized EV charging system.

ordinated EV charging occurs at home, particularly from evening to early morning or whenever the user arrives home. The power flows from the grid to vehicle G2V in a traditional one-way direction.

A large number of recurring uncoordinated charges would deteriorate the distribution system's functioning [i.e., transformer overloading, voltage instability, power loss, and frequency variation], which may collapse the power system [18,19,57,58,62,63]. Figs. 3 and 4 show the impact of uncoordinated home EV charging on the residential aggregate electricity demand considering different PEV market shares, including a reference scenario with no EVs [94]. However, a smaller portion of EV penetration has no significant impact on the grid side [95]. The potential advantages of uncoordinated EV charging are 1. No communication infrastructure is needed. 2. It is economical. The charging components are cheaper than the level-2 and level-3 charging components. 3. Need for a third party is neglected, and hence no risk of privacy issues.

3.2.2. Coordinated EV charging

In general, an appropriate management scheme is required for optimal EV charging. Coordinated charging will enhance the operational performance of the grid through intelligent charging management. In coordinated charging, the whole operation is controlled directly by a third-party controller or aggregator. The aggregators and controllers are responsible for achieving the objectives, such as cost minimization, power loss minimization, and others. Henceforth, the EV users and private and public charging station operators are involved in charging management by selling or buying electric power. Therefore, the power flow between the EV and grid can occur in unidirectional and bidirectional modes.

Fig. 5 represents EV charging without an aggregator, and Figure 6 depicts EV charging with the aggregator.

3.2.2.1. Unidirectional mode. In the coordinated unidirectional mode charging scheme, the charging environment is technically upgraded compared to the conventional charging environment, i.e., chargers used are capable of two-way communication. Therefore, the controller or aggregator exclusively controls the charging management with the help of an intelligent grid environment [61,68,70,74]. Monitoring the EV charging avoids grid overloading and undesired impacts. The EVs are charged during off-peak hours or while the grid has surplus power to achieve a flat load curve. The advantage of the unidirectional mode is that minimal communication infrastructure is required. The level of complexity is less when compared with the bidirectional charging mode, and the active participation of users in the energy market is enabled. The drawback of the system is that it could not guarantee the optimal global solution as the nonexistence of global information [96].

3.2.2.2. Bidirectional mode. Bidirectional charging allows the battery to charge and enables the battery to supply the energy back to the grid with some constraints to manage the energy spike in the grid. Hence, the bidirectional EV chargers can facilitate electrical energy charging and discharging. In this mode, the users must agree with the aggregator for controlling the charging and discharging operation and as well as to provide ancillary services such as frequency control, voltage regulation, etc. Many research works have shown the potential benefit of V2G/V2B [97–100]. However, frequent charging and discharging would reduce the battery life, and a customer would not be interested in replacing

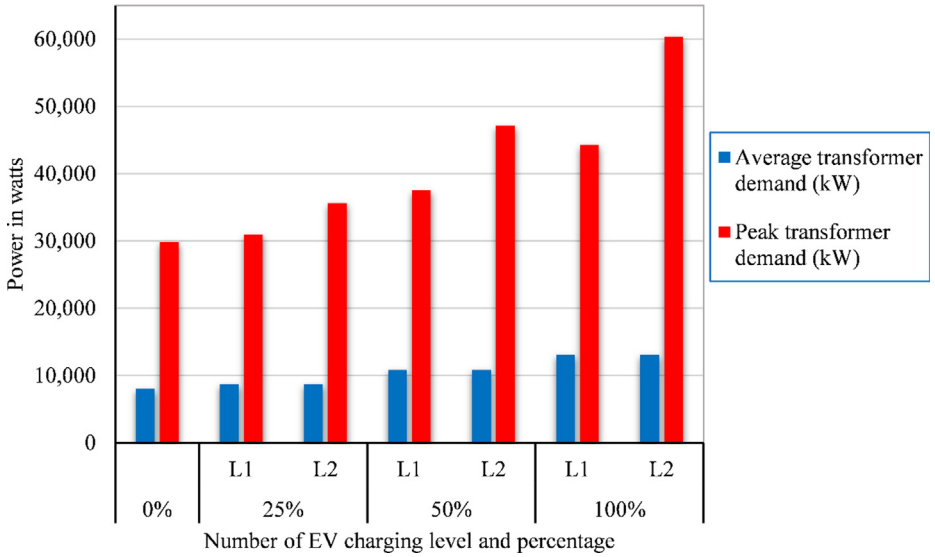


Fig. 3. Influence of uncoordinated PEV charging (in-home) on electricity demand for 200 households. L1 denotes level 1 charging, and L2 denotes level 2 charging [94].

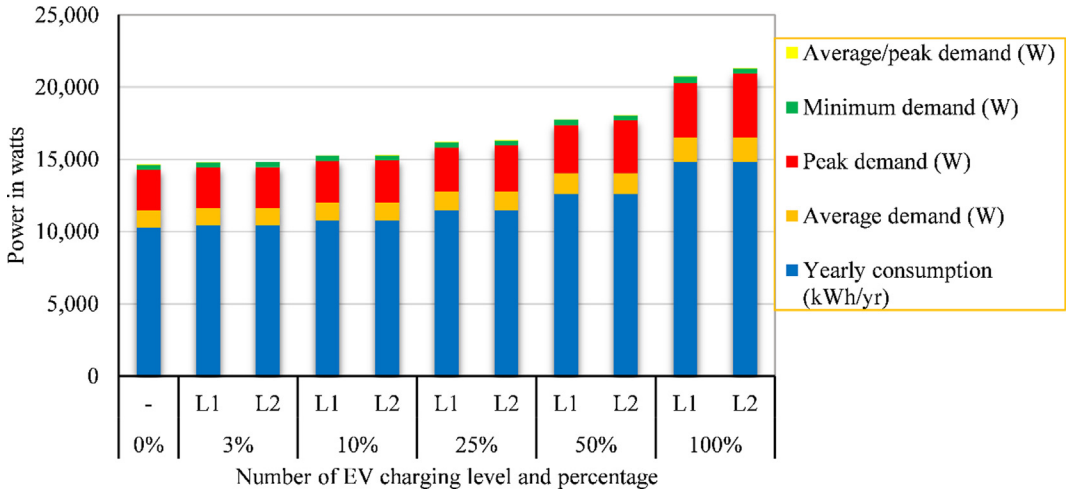


Fig. 4. Influence of uncoordinated PEV charging (in-home) on electricity demand for a distribution transformer connected to 6 households. L1 denotes level 1 charging, and L2 denotes level 2 charging [94].

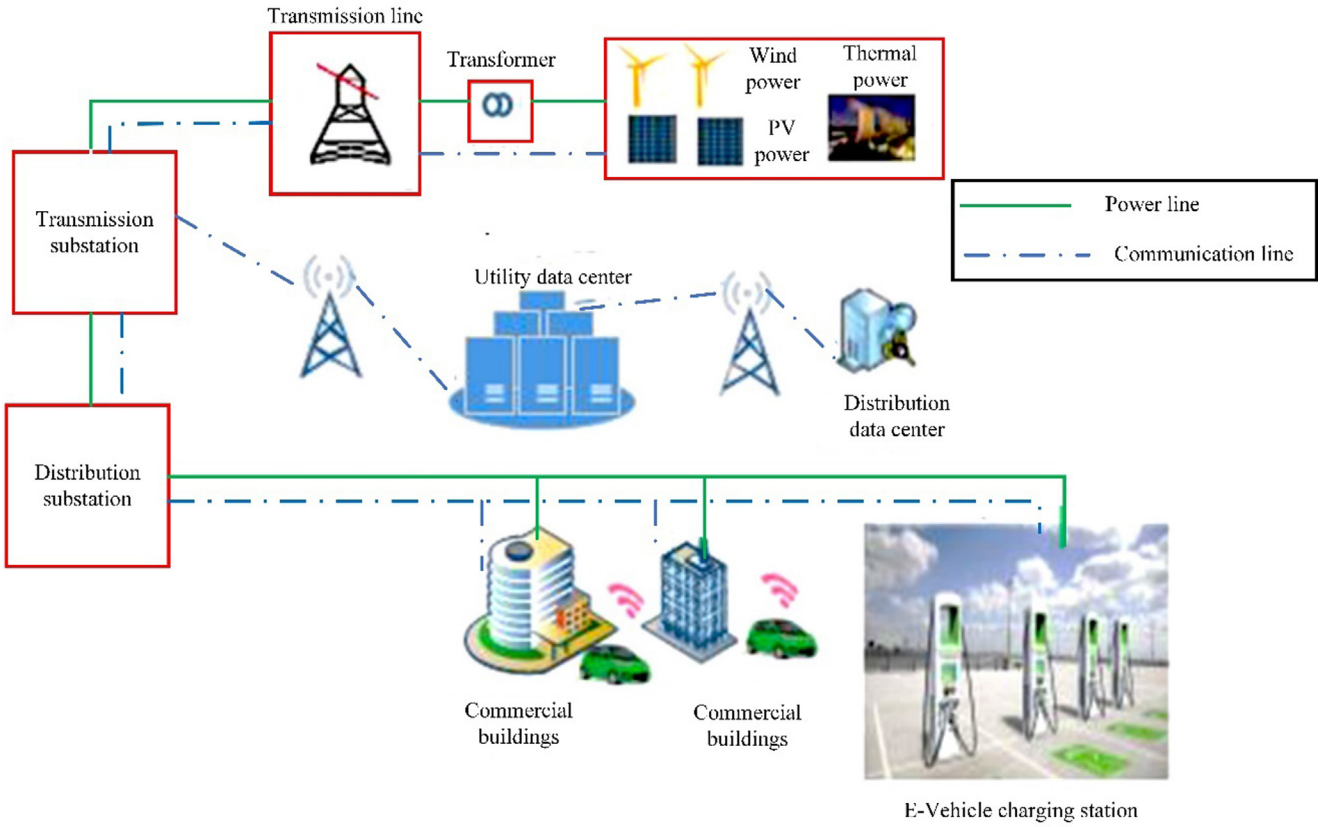


Fig. 5. Uncoordinated EV charging.

the battery frequently, as it is not economical [101]. In addition, it requires a greater communication facility, and the information flow would be complex. Hence, it is time-consuming to produce a globally optimal solution [102–104]. A comparison of coordinated and uncoordinated charging is given in Table 3. In addition, a few reviewed papers are compared concerning different power levels, support of RES, aggregators, etc., which are presented in Table 4.

4. Charging station identification

One of the critical concerns of an EV user is the driving distance covered by the EV battery [105]. So, whenever the EV's battery drains, the EV user must find a Charging Station (CS) to charge the EV [106]. Currently, most EV users depend on a third party (aggregator) to reserve a charging slot to charge the EVs. The aggregators collect the details of the EVs from the user to provide the

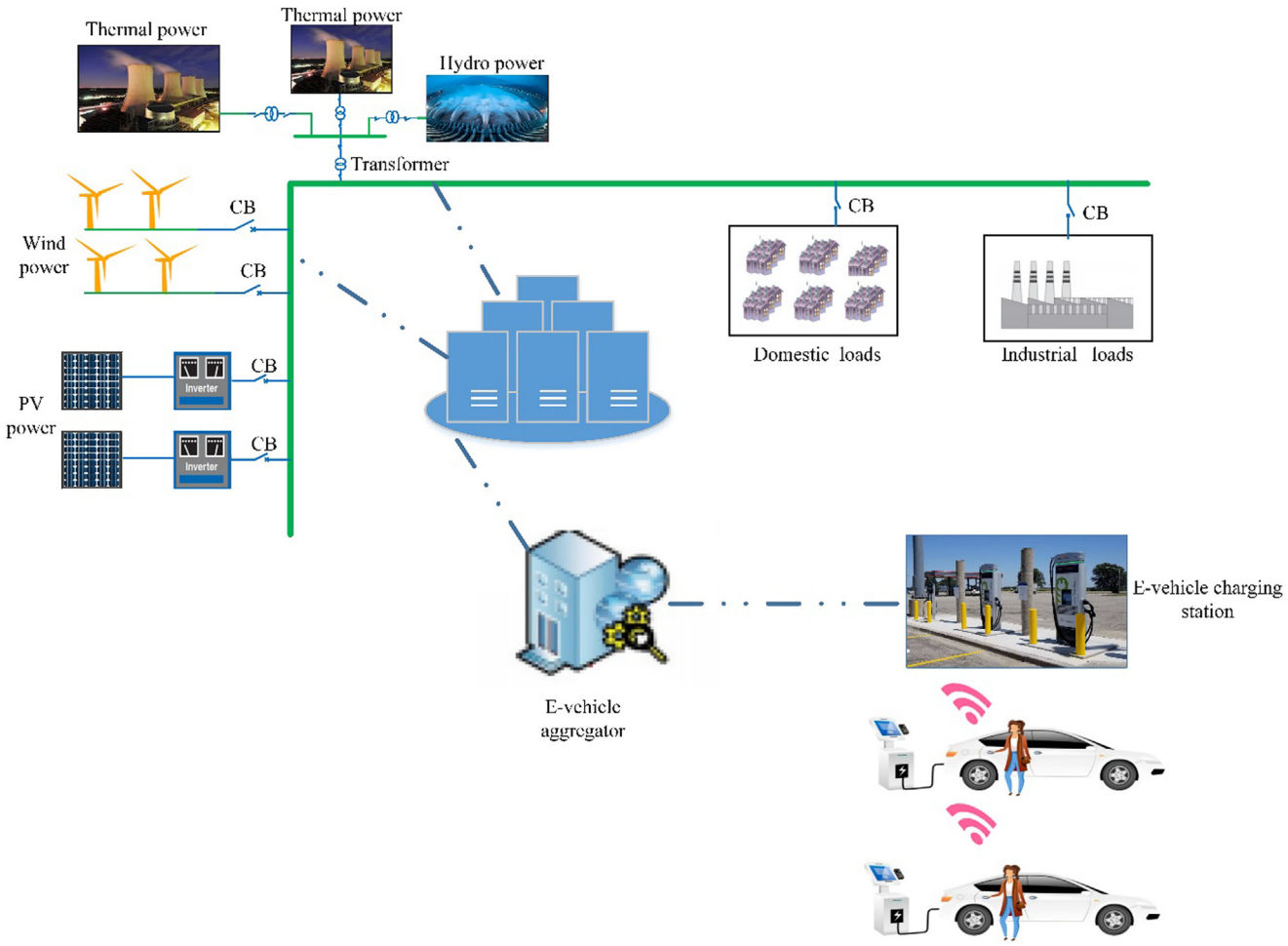


Fig. 6. EV charging with the aggregator.

Table 3
Comparison of coordinated and uncoordinated charging.

	Uncoordinated	Coordinated	
		Unidirectional	Bidirectional
Charging decision	User	Aggregator/controller	Aggregator/controller
Control	User	Aggregator/controller	Aggregator/controller
RES adoption	Partial	Partial	High
Ancillary support	Partial	Partial	High
Flexibility level	Low	Medium	High
Communication infrastructure level	Not required	Medium	High
Information sharing	Not required	Easy	Complex
V2G/V2H support	Not available	Not available	Available
Infrastructure cost	Low	Average	High
Beneficiary	User	All the participants	All the participants

service [40,41,45,48,54–58,61–63]. However, not all the customers may be interested in sharing their information, such as phone numbers [107–109]. At least fifty percent of customers are highly concerned about their privacy, and the other fifty percent are worried about their confidentiality based on the situation. The aggregators depend on other service providers (i.e., independent system operators, independent power producers, etc.). Hence, the information exchange between the user and service provider becomes more complex. It makes the customer spend more cash on the charging of EV, particularly at peak hours. However, knowing the real-time status of the CS (such as the location of CS, charging slot availability, charging cost, etc.) without an aggregator is more valuable to the consumers. Besides, it is beneficial to the CS operator to make a charging schedule in advance.

4.1. Influence of third-party/controllers/aggregators in EV charging

In general, the PLs are operating with the First in First Serve (FIFS) method. However, charging by the FIFS method may not minimize the electricity purchase cost due to the hourly changing electricity price. So, the optimal scheduling of the EV is to be carried out to reduce the electricity purchase cost of the PL operator. If the PL/EV charging station operator knows the arrival time, departure time, the demand of the EV, etc., the scheduling may

Table 4
Comparison of articles reviewed in this work.

Ref No	Power Level			RES/ESS support	Power flow		Aggregator/Controller	Type		
	1	2	3		Unidirectional	Bidirectional		Residential	Private	Public
[58]	✓	✓	×	×	✓	×	×	✓	×	×
[59]	✓	×	×	✓	×	×	×	×	×	×
[68]	✓	×	×	×	✓	×	✓	✓	×	×
[69]	✓	×	×	✓	×	✓	×	✓	×	×
[70]	×	✓	×	×	✓	×	✓	✓	×	×
[71]	✓	×	✓	×	✓	×	×	✓	×	×
[72]	✓	✓	✓	×	✓	×	×	✓	×	✓
[73]	✓	✓	✓	×	✓	×	×	×	×	✓
[74]	✓	✓	✓	×	✓	×	✓	✓	✓	✓
[75]	✓	×	✓	✓	×	✓	✓	✓	×	✓
[76]	✓	✓	✓	×	×	✓	✓	✓	×	✓
[77]	×	✓	×	×	✓	×	✓	✓	✓	✓
[60]	✓	✓	✓	×	×	✓	✓	✓	×	×
[78]	✓	✓	✓	×	×	✓	✓	✓	✓	✓
[79]	✓	✓	✓	×	✓	×	×	✓	✓	✓
[80]	×	×	✓	×	×	✓	×	×	×	✓
[81]	×	✓	×	×	✓	×	✓	×	×	✓
[82]	✓	✓	✓	✓	×	✓	✓	×	×	✓
[83]	×	×	✓	✓	×	✓	✓	×	×	✓
[61]	✓	×	×	×	✓	×	✓	✓	×	×
[62]	✓	×	×	×	✓	×	×	✓	×	×
[63]	×	✓	×	×	✓	×	×	✓	×	×
[64]	×	✓	×	✓	×	✓	×	✓	×	×
[65]	✓	✓	✓	×	×	✓	✓	✓	×	✓
[66]	✓	××	×	×	✓	×	×	✓	×	×
[67]	✓	×	×	✓	✓	×	×	✓	×	×
Proposed	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

be done effectively. But the infrastructure available for users to book a charging slot as they wish is not developed effectively.

At present, EV users are charging their EVs through third-party aggregators. Aggregator acts as an agent between electricity producers and EV customers and represents a group of customers who responds to them with an agreement. Each aggregator must provide the requested power to the EV customers. Besides, they can install communication and control devices (i.e., smart meters) at the customer end. They communicate with the electricity companies on behalf of their customers on the prices and power demands. This leads to customer privacy issues as the users must share all the details.

The main concern about EV charging is not disclosing the user's details such as name, address, phone number, e-mail address, and banking details. In [110], EVs usually request the global aggregator to make the charging decision. Throughout this process, the information on the EVs is released as anticipated. The privacy of the EV user is necessary, as a harmful trader may disturb an individual EV by marketing manufactured goods or services. The EV user may not know whether CS is available nearby or not, which may lead to travel to the CS located far away from the EV location. A simple mobile application is proposed in [111] to show the list of CSs based on the vehicle's geographical location. This application provides three essential information to the user: CSs near the EV, travel distance to a particular CS, and the time to reach the CS. But this application is not able to find the actual status of the CS to book a slot. In [112], the CS suggestion scheme and the EV battery's SOC calculation. The suggested system notifies the vehicle owner whenever the SOC of the battery reaches the pre-set value. But the investigated method does not give the location of the CS and the availability status of the CS.

In [113], a CS suggestion scheme for EV taxis is developed. This method can calculate the total travel time to the CS and recommend the nearest CS. However, the proposed system cannot provide the actual status of the CSs to book a charging slot. In [114], a charging scheduling algorithm is proposed to minimize travel time using a game-theory model. The developed algorithm also reduces the EV charging time and the idle time of the charging point. In [115], a traction power estimation model is used for CS suggestions to the EV users. A range indicator shows the distance covered by the EV with the power left in the battery. Although the system indicates the range, this method could not provide the actual status of the CS. In [116], an activity-based model is developed to analyze the queuing delay in CSs and provides decision support for the EV user to select a CS. It is assumed that the charging option for the EV user decreases whenever a CS is not chosen in the drive.

Further, this method is based on numerical calculation and doesn't provide the CS status to the user. Based on the same activity-based model, a method is proposed in [117] to find and select the CS to minimize the range anxiety of the EV user. However, this method suggests only the location of the CS, and it doesn't provide any option to reserve the charging slot.

A mobile-based Vehicle to Anything (V2A) application system is developed in [118] to help the EV user. The developed application alerts the driver about the remaining power left on the battery, the distance covered, the location of the CSs, the route to the CSs, etc. In [119], a trip-planning scheme for the EV driver is formulated by optimizing the trip parameters. A driving pattern is suggested to optimize the battery efficiency to minimize range anxiety. However, the EV user behavior determines the vehicle's speed. But the user behavior differs from user to user.

Further, the proposed method does not consider the location of the CS, route to the CS, and actual status of the CS. An approach is developed in [120] to reduce the stress on range anxiety based on the driver's behavior. The characteristics of the EV drivers are analyzed from their trips. This approach considers various factors such as vehicle type, mileage of the vehicle, etc., are considered in this approach. A model is developed for optimal charging by considering the CS queue and EV demand [121]. A price stimulus method is developed to assist the user in selecting the CS to minimize the charging cost. However, it is assumed that the user selects the CS nearer to their location. It makes the user wait for more time, creating congestion in the CS.

In [122], GA is used to optimize the charging routes for EVs. By using this method, a user can be directed to a CS, which is already congested. Therefore, developing an application to provide the exact status of the CSs in a particular region is a high-priority-one. The charging station recommendation models are getting attention for a wide acceptance of EVs in the market. Some research works [84,88–100] recommend EV charging through an aggregator and finding an optimal CS based on mathematical formulations. Though a real-time server-based application is developed to identify a suitable EVCS for the user [33], the complexity may increase when the number of EVs increases. The variation in EV charging cost may affect the user choice, as most users are likely to charge their EV when the price is low. In addition, suggesting the charging slot accessibility in various CSs would reduce the congestion. Hence, an inadequate number of CSs can impact load forecasting and power distribution for the customers. Accordingly, as shown in the figure, a server-based real-time application can be used for EV scheduling at CSs and to intimate the actual status of CSs to the EV user. The real-time status of CSs includes charging cost, location and distance of CS, reserve, fast charging availability, etc. The benefits of the recommended scheme related to other techniques conferred in the literature are given in Table 5.

4.2. Challenges and proposed recommendations

Adopting EVs on the market benefits the electric power grid and reduces the air pollution by the ICE vehicles. From the air pollution point of view, the EV emits very little or no emission. From the power network point of view, it provides many opportunities, such as V2G, V2B, Smart metering, etc., and facilitates the integration of sustainable energy resources. However, there is a wide range of challenges and barriers to be rectified for sustainable market growth.

4.2.1. Challenges

The main concern of EV users is the range anxiety determined by the battery. At present, the battery technologies are not at a mature level. Many batteries have a limited life cycle and high initial cost and are not flexible with the varying temperature. From an economic point of view, the cost is an important parameter that restricts EV penetration in the market. The cost involved with the individual charging point, battery replacement cost, the cost of communication, and charger cost are high, and they are not suitable for an average-earning family.

Further, most of the charger technologies available in the market support one-way power transfer (Grid to Vehicle), and the development of bidirectional chargers are at an emerging stage. From an infrastructure perspective, the available charging infrastructure is inadequate to meet the user demands. A reliable communication network is required for the participants to communicate with each other. An unreliable network will result in chaos. There is a need for a proper government policy to enable smart grid technology. Because most customers depend on a third

Table 5
Comparison of the research works reviewed in this work.

Ref.	Year	Goal	Tool/Method	Tariff	EV*						Inference
					1	2	3	4	5	6	
[60]	2020	To provide regulation service in a residential PL	Matlab	ToU	✓	×	✓	×	×	×	From the user input, the aggregator provides the regulation service for both slow and fast charging cases.
[61]	2014	To minimize the charging cost for the user	Matlab	ToU	✓	×	✓	×	✓	×	Aggregator or DSO influences the EV charging for cost minimization
[65]	2021	To make benefit market participants by optimal scheduling	Matlab	ToU	✓	×	✓	×	✓	×	DR program is introduced to benefit the user, aggregator, and DS. The user is dependent on aggregators for EV charging.
[68]	2019	To reduce the charging cost through optimal scheduling	Matlab	ToU	✓	×	✓	×	✓	×	The aggregator knows the charging time. Hence, the scheduling is made to reduce the cost.
[74]	2019	To reduce the computational complexity of EV charging	Python-based convex optimization toolbox	N/A	✓	×	✓	✓	×	✓	A pre-agreement is made between aggregator and user for EV charging and the aggregator depends on the global controller.
[75]	2020	To support charging without enhancing the infrastructure	Matlab	N/A	×	×	×	×	××	×	The aggregator depends on DSO for different load demands, so the user must change the schedule.
[76]	2020	To make optimal charging decisions considering user convenience.	Matlab	N/A	✓	×	×	✓	×	×	The central controller is responsible for EV scheduling, and the local aggregator collects user data.
[77]	2014	To minimize the operational cost through charging/discharging	Matlab	N/A	✓	×	✓	✓	×	×	The operational cost is minimized by reserving the charging and discharging slots for EV users.
[78]	2020	To make use of the user's flexibility for optimal charging/discharging schedule.	Matlab	N/A	✓	×	✓	✓	✓	×	The frequent charging/discharging degrades the battery life, and this will affect the user's convenience.
[112]	2014	To support the EV users by monitoring the battery status and providing a dynamic range prediction.	GPS	TOU	×	✓	✓	✓	×	×	The proposed scheme does not support the user by providing the status of CSs. The user may not know whether the CS is currently functioning or not.
[113]	2015	To reduce the user's concern on range anxiety.	MS excel	N/A	×	✓	✓	✓	×	×	This model is developed using offline data. The proposed scheme calculates the range of the EV with power left in the battery.
[115]	2017	To develop a CS recommendation scheme based on a range indicator.	Traction power estimation	N/A	×	✓	✓	×	×	×	A range indicator using the SOC calculation model is developed to alert the EV user about the range. It assumes that the system can book a charging slot in a CS.
[52]	2020	To provide real-time charging station information	Web (cloud)	TOU	×	✓	✓	✓	✓	✓	Information sharing between CSs, and users will be of benefit to both the user and the CSs. Customer-oriented, user-friendly.

* 1: Aggregator intervention, 2: EV user privacy, 3: CS location information, 4: User preference, 5. Load forecasting recommendation, and 6: Hassle-free information exchange network.

party to charge their EVs, the customer's privacy is to be mainly focused as discussed in the paper.

4.2.2. Proposed recommendations

A comprehensive research analysis is required around battery technology. Because the battery is the major component of the EV. The government can introduce policies to provide incentives for using EVs. Though many countries have announced tax-free registration, the EV user must be provided with some additional credits such as a discount in income tax, a discount on electricity bills, etc. Further, it is necessary to assess the impact of issues related to user privacy. Hence, suggestions are provided to make the EV charging in a protected manner without an aggregator or controller. A real-time CS recommendation application is helpful to the user to share the status of CSs, such as charger option, energy cost and the distance to reach the CS, etc. It is represented in Fig. 7.

The application should collect the EVs and CS status, such as the power available in the EV battery, location of EV, location of CS, charging cost, distance from the EVs location, fast charging option, charging slot availability, etc. Such an application could help the user overcome drawbacks like long queuing at CSs and third-party involvement in charge the EV. It would also help the EV user find a suitable CS and the CS operator make an optimal schedule for charging the EVs. It must be capable of preventing the complicated information exchange between the service provider and the customer. A novel application that can utilize the benefit of the Internet of Things (IoT) is required to overcome these issues. A reliable communication network is to be provided by the government or from public agencies to the users. This would be more beneficial to all the participants involved in the market to avoid concern about privacy and save money and time for charging the EV at the CSs.

5. Conclusion

The development of alternative energy sources through investment and technical innovation is gaining traction due to growing

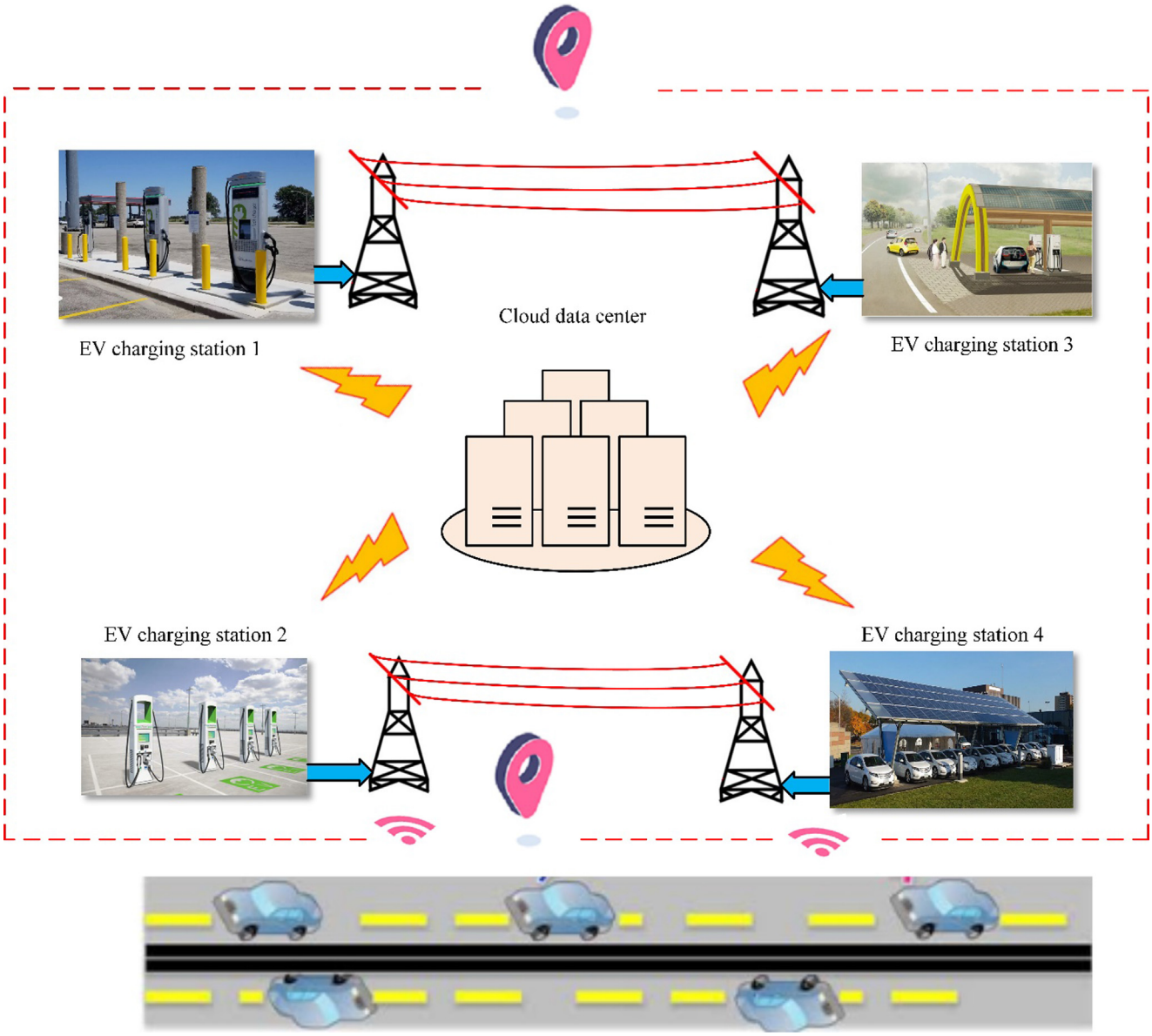


Fig. 7. EV charging station information sharing system.

awareness of global warming and climate change as well as efforts to lessen reliance on fossil fuels as the major energy source. A significant portion of the world's power consumption is used for transportation, a key economic engine. Due to its intrinsic capacity to have a beneficial impact by lowering dependence on fossil fuels and carbon emissions, EVs are set to become a popular form of transportation. Government legislation in several nations and technological and economic improvements made this possible, which increased the number of EVs on the road. In this paper, an attempt to review the works that have been done on EV charging technology, infrastructure, issues related to charging station identification, and CS recommendation techniques in the past decade is carried out.

Several EV charging technologies are examined in depth. Although DC fast-charging stations are more expensive to build than AC charging stations, several automakers are working to establish DC fast charging stations since the latter can charge EV batteries more quickly. Additionally, the EV consumer's needs are considered when choosing the types of CSs. For the range of 0–10 km in the domestic outlets, a level 1 charging station is selected. For the 50 km range, a level 2 charging station was used. For the 100 km range, a level 3 charging station was used. The 100–200 km range is chosen for level 4 of the DCFC station.

The second trend of charging infrastructure was analyzed in detail. The development of charging infrastructure has gained significant attention in developed and developing countries as inadequate charging infrastructure can affect EV promotion in the market. The charging infrastructure comprises power and a communication system.

Another critical concern of an EV user is the driving distance covered by the EV battery. So, whenever the EV's battery drains, the EV user must find a CS to charge the EV. Therefore, the suggestions for the user-friendly selection of charging stations are discussed in detail.

Hence this study will assist in understanding the most important relevant information about the existing works. Though the recent development in charging technology and infrastructure supports the adoption of EVs in the market, the following knowledge gaps can be considered.

1. The influence of economic factors on EV customers, such as charger and charging and battery costs, has not been considered in real-time.
2. Additional requirements such as smart communication systems for intelligent charging, reliable charging systems, IoT enabled CSs are not well developed. This will also help further research EV charging and infrastructure development related to intelligence systems.
3. Customer privacy is still a big challenge. Therefore, extensive research would be needed to avoid privacy issues.

These are the key areas to be focused on in future works to achieve a sustainable transport system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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